

Project Plan Draft

Project Title

Automatic Classification of medical X Ray images

Description of the project

Classifying medical images has been heavily researched and trialled, as being able to do so effectively would mean diagnoses would be able to be done much quicker. This would lead to much better patient outcomes. One approach to this, is to use a Convolutional Neural Network (CNN), which learn features from images, and use these to accurately classify images into the desired classes or categories.

However, one problem with this, particularly when it comes to medical images, is that the classes may be heavily imbalanced, due to different parts of the body being X rayed more than others, and certain diseases being x rayed more, as, for example, they were discovered earlier. This means that, in a given dataset of chest x rays for example, there may be many more images labelled as “pneumonia”, compared to other classes such as “Lung Opacity”, which would mean that the model may be “biased” towards the majority class. This means it may inaccurately label images as “pneumonia”, which, in the medical context, could have serious medical consequences.

This project aims to design, develop and evaluate a robust convolutional neural network to classify chest x ray images, despite this class imbalance, to properly classify unseen chest x rays. The dataset I have chosen for this, is the ChexPert Small dataset from Stanford ML. The ChexPert small dataset contains 224,316 chest x ray images, split into 14 classes. The dataset is also quite imbalanced, with 46,000 images labelled as “lung lesions”, whereas over 100,000 were labelled as “lung opacity”, making this dataset a good candidate to investigate the effect of class imbalance on model performance.

This project does not come without its challenges, for example, the issue of how exactly I address the problem of class imbalance is crucial to the performance of my model. I will need to use augmentation techniques such as rotating, scaling, and changing contrast to create a more diverse dataset, and will need to consider other techniques such as undersampling to balance the datasets. Another challenge will be trying to find the right balance of complexity, and computational performance when training the model. A more complex model, with more hidden layers of convolution and pooling layers, may lead to better accuracy and a lower false classification rate, however, I will need to balance this with the physical constraints of the computing power I have available. I will also need to consider the methods with which I evaluate the model – for example, accuracy, while an okay general indicator of performance, can be misleading, as the model could get a high accuracy score by just predicting the majority class, regardless of what the actual image shows.

Main technical problem of the project

The main problem that my project aims to solve, is to design, develop, and evaluate a robust CNN model which can accurately classify different chest x ray images, despite the significant class imbalance present in the dataset. If class imbalance is not properly addressed when developing an x ray image classifier, it could lead to the model being biased towards the majority class. This would mean that there is a higher chance of misdiagnosis, which is serious issue if it happens in the medical field. The project will consider all aspects, from data preparation, to the complexity and

depth of the model itself, and the use of a variety of evaluation metrics to create a robust, accurate medical x ray classifier, to help with diagnosing certain conditions quicker.

Indicative Timeline (all dates are approximate)

October 16 - Nov 10th:

- Research imbalance further – ways to deal with imbalance – e.g. augmentation
- Research implementation methods, libraries etc, decide on methodology

Nov 10th - Nov 20th:

- write up interim report

Nov 21 - ~Dec 10th

- Implement and test baseline model

~Dec 11th - February 20th:

- Iteratively improve the model, adding and experimenting with different augmentation techniques (e.g. rotation, scaling, adjusting contrast etc), changing number of layers, experimenting with different optimization functions
- Implement the evaluation algorithms (e.g. F1 Score, Precision, Recall) and evaluate the model

9th february - 20th Feb

- interview preparation
- attend interview

March – April 25th:

- Write dissertation, and improve it over time
- Prepare viva/presentation
- Final touches on software system

April 25 to April 27th:

- Check all components and get them ready for submission

April 27th:

- Submit all necessary components